

Adiabatic flame
temperature

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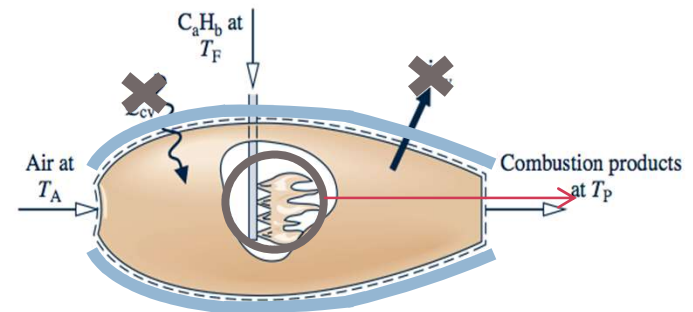
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Adiabatic flame temperature

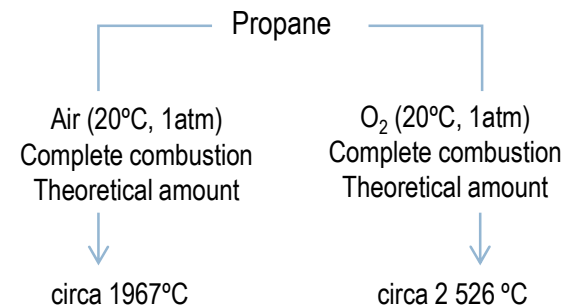
- › In a open system energy is transferred through:
 - Mass, Work and Heat.
- › If $W_{\text{syst}}=0$, only through products and heat trans
 - Therefore:
 - › Smaller Q, higher T_{prod}

- › Adiabatic flame temperature:
 - In an adiabatic system:
 - › Product temperature reaches maximum
 - › Energy is only used to heat the products.
 - › Also known as Adiabatic combustion



Adiabatic flame temperature

- › The T_{ad} is specific for a process, it depends of:
 - › The type of fuel
 - › Temperature and pressure of the reactants
 - › Rate of combustion – complete or incomplete
 - › Amount of air/oxidant used.
- › Maximum T_{ad} ($T_{ad,max}$) value for a specific process:
 - Complete combustion with theoretical amount of air (ideal)
 - › Specific process: certain fuel, with reactants at certain state.
 - The real measured T_{ad} value is several degrees below $T_{ad,max}$
 - › **Why do you think this happens?**

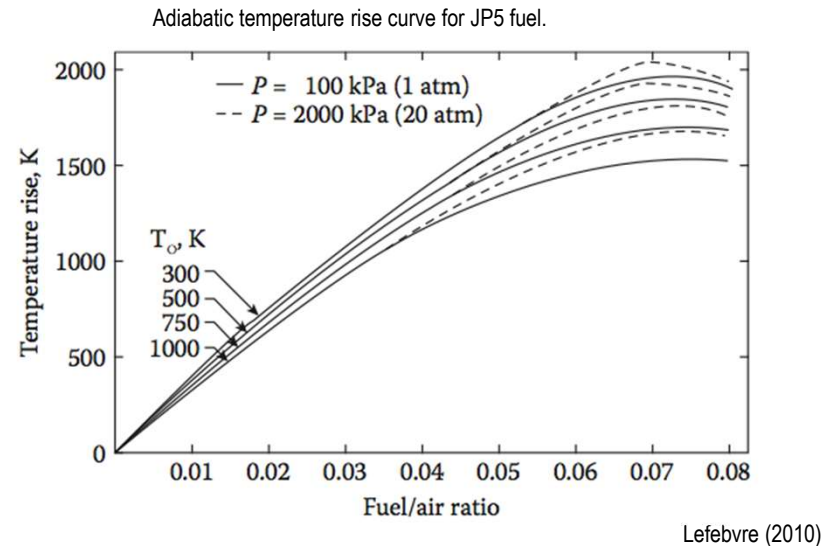


What affects the max t_{ad} value?

- › Heat losses occur – no system is 100% adiabatic:
 - Conduction, radiation and convection
- › Excess of air– there is an adequate amount
 - Functions as a diluent.
 - › Also, ideally, nitrogen doesn't react but its heated
 - High excess aids dissociation processes
- › Dissociation:
 - At high temperatures molecules break down
 - At low T – CO_2 , H_2O , N_2 At high T – CO , H_2 , $\cdot\text{O}$, $\text{H}\cdot$, $\cdot\text{OH}$
- › State of reactants
 - Heating and phase changes implies energy consumption

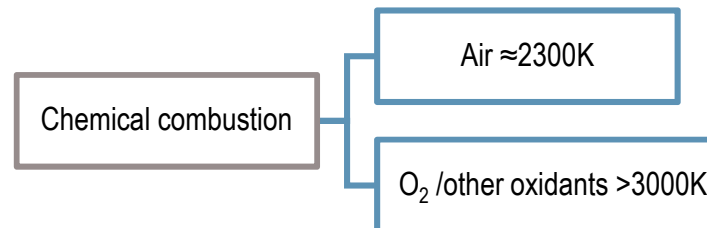
Important variables

- › $AF_{\text{ratio}} / FA_{\text{ratio}}$:
 - The combustion is complete or incomplete
- › Temperature air
 - Increase in T_{air} , increases T_{ad}
 - › No energy losses from heating.
- › Pressure
 - For a constant T_{air} , increase pressure increase T_{ad}
 - › High T , increases P and degree of dissociation of products
 - › This has an effect over system volume
 - › Constant volume, pressure increases, dissociation reduces
- › Air vitiation
 - Precombustion: CO_2 and H_2O



Importance of T_{ad}

- › Maximum process temperature:
 - Limits the process
 - › Rarely a real process reaches this temperature
 - Calculates pollutants amount
- › Estimates thermal effects:
 - Considerations for materials
 - Considerations in design, handle and security of process
- › Important for combustion efficiency.



Determining T_{ad}

- › Calculate $H_{\text{reactants}}$
 - Quick and direct
 - Their state is known
 - Data can be found in tables
- › Determining H_{products} is different
 - $T_{\text{prod}} = T_{\text{ad}}$ is unknown
 - Can be calculated by:
 - › Enthalpy balance iteration
 - › Average specific heat (C_p)

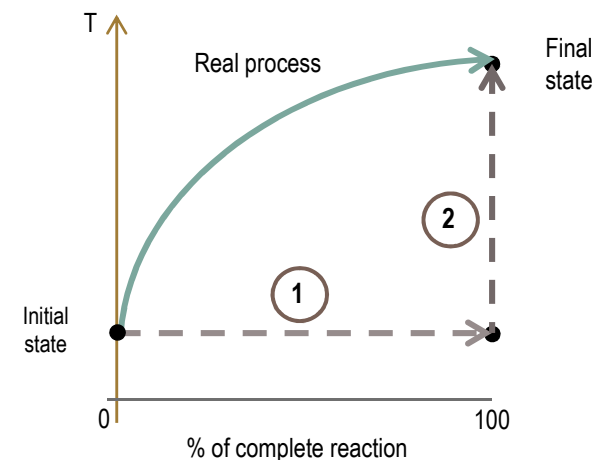
$$Q - W = H$$

If, there is no work done
And E_k, E_p negligible

$$Q = \sum (N_i \bar{h}_i)_R - \sum (N_i \bar{h}_i)_P$$

$$\sum (N_i \bar{h}_i)_P = \sum (N_i \bar{h}_i)_R$$

$$\sum N_P (\bar{h}_f^\circ + \bar{h}(T) - \bar{h}^\circ)_P = \sum N_R (\bar{h}_f^\circ + \bar{h}(T) - \bar{h}^\circ)_R$$



Method 1: Enthalpy balance

- › General equation for open system:

$$\sum_{\text{T}_F \text{ of the products} \rightarrow \text{T}_{ad}} N_P (\bar{h}_f^\circ + \bar{h}(T_{ad}) - \bar{h}^\circ)_P = \sum N_R (\bar{h}_f^\circ + \bar{h}(T_F) - \bar{h}^\circ)_R$$

- › Re-arranging the terms:

$$\sum N_P (\bar{h}(T_{ad}) - \bar{h}^\circ)_P = \sum N_R \bar{h}_f^\circ - \sum N_P \bar{h}_f^\circ + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R$$

$$\sum N_P (\bar{h}(T_{ad}) - \bar{h}^\circ)_P = -q_{rsn} + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R$$

- › Depending of the scenario, we have to make supposition on the temperature we are calculating

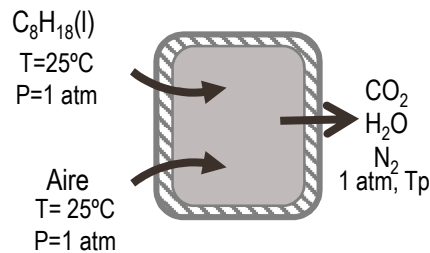
$$\sum N_P (\bar{h}_f^\circ + \bar{h}(T_{guess}) - \bar{h}^\circ)_P = \sum N_R (\bar{h}_f^\circ + \bar{h}(T_{guess}) - \bar{h}^\circ)_R$$

- › We will iterate with the unknown temperature until the energy balance is satisfied:

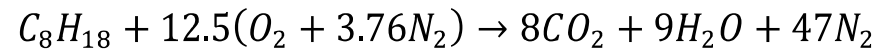
$$\sum N_P H_P = \sum N_R H_R$$

- › Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm.

Suppositions: 1) open system, steady-state; 2) No work is done; 3) All gases are ideal gases



Chemical reaction balanced:



Energy balance:

$$\sum N_P \bar{h}_p(T_{ad}) = \sum N_R (\bar{h}_f^\circ)_R - \sum N_P (\bar{h}_f^\circ)_P + \sum N_P (\bar{h}^\circ)_P$$

$$\sum N_R (\bar{h}_f^\circ)_R = -249950 \text{ kJ/kmol}$$

$$\sum N_P (\bar{h}_f^\circ)_P = -5\,324\,540 \text{ kJ/kmol}$$

$$\sum N_P (\bar{h}^\circ)_P = 571\,491 \text{ kJ/kmol}$$

Enthalpy of formation	
	$h_f^\circ \text{ (kJ/kmol)}$
C_8H_{18}	-249 950
O_2	0
N_2	0
$H_2O_{(g)}$	-241 820
CO_2	-393 529

In order to calculate the sensible enthalpy of the products, a final temperature of the outlet must be known – or supposed.

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Supposed T = 2500 °K

$$\sum N_P(\bar{h}(T_{2500K}))_P = 5\,930\,239 \text{ kJ/kmol}$$

$$5\,930\,239 \text{ kJ/kmol} = 5\,646\,081 \text{ kJ/kmol} \quad \text{Our guess give us a higher value}$$

Supposed T = 2000 °K

$$\sum N_P(\bar{h}(T_{2000K}))_P = 4\,595\,839 \text{ kJ/kmol}$$

$$4\,595\,839 \text{ kJ/kmol} = 5\,646\,081 \text{ kJ/kmol} \quad \text{Our guess give us a lower value}$$

Extrapolate or interpolate:

Give us a Tad = 2 392.87°K

H at different temperatures (kJ/kmol)

T (°K)	CO ₂	H ₂ O	N ₂
298	9364	9904	8669
2000	100804	82593	64810
2200	112939	92940	72040
2500	131290	108868	82981
2700	143620	119717	90328
3000	162226	136264	101407

Method 2: $C_{p,i,avg}$

- It's also based in a energy and mass balance

$$\sum N_P (\bar{h}(T_{ad}) - \bar{h}^\circ)_P = \sum N_R \bar{h}_{f_R}^\circ - \sum N_P \bar{h}_{f_P}^\circ + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R$$

- Based on the supposition that we can approximate the term of sensible enthalpy with an average specific heat function:

$$\sum N_P (\bar{h}(T_{ad}) - \bar{h}^\circ)_P \cong (T_{ad} - T_0) \sum N_P \bar{C}_{p,avg,P}$$

$$(T_{ad} - T_0) \sum N_P \bar{C}_{p,avg,P} = \sum N_R \bar{h}_{f_R}^\circ - \sum N_P \bar{h}_{f_P}^\circ + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R$$

- We can solve for our unknown variable:

$$T_{ad} = T_0 + \frac{\sum N_R \bar{h}_{f_R}^\circ - \sum N_P \bar{h}_{f_P}^\circ + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R}{\sum N_P \bar{C}_{p,avg,P}}$$

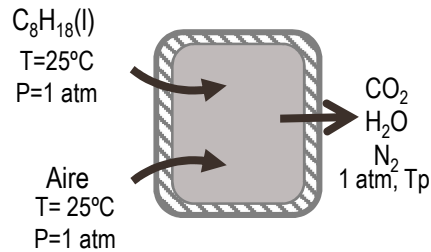
- We have the variable average specific heat

$$\bar{C}_{p,avg,P} \text{ is } \bar{C}_{p,P} \text{ @ } \bar{T} \quad \bar{T} = 0.5(T_{prod} + T_{react})$$

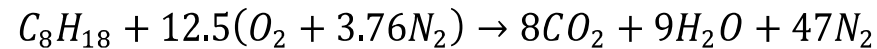
- There are tabulated data for specific heat at different temperatures and for a variety of compounds.

- Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm

Suppositions: 1) open system, steady-state; 2) No work is done; 3) All gases are ideal gases



Chemical reaction balanced:



Based on the energy balance:

$$T_{ad} = T_0 + \frac{\sum N_R \bar{h}_f^\circ - \sum N_P \bar{h}_f^\circ + \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R}{\sum N_P \bar{C}_{p,avg,p}}$$

$$\begin{aligned} \sum N_R (\bar{h}_f^\circ)_R &= -249950 \text{ kJ/kmol} \\ \sum N_P (\bar{h}_f^\circ)_P &= -5\,324\,540 \text{ kJ/kmol} \\ \sum N_R (\bar{h}(T_F) - \bar{h}^\circ)_R &= 0 \end{aligned}$$

Calculating $C_{p,avg}$, @ T_{avg} :

- Suppose a T_{ad} value
- Obtain T_{avg}
- Obtain $C_{p,avg,p}$ @ T_{avg} from table of C_p

$$T_{ad,guess} = 3\,000^\circ K$$

$$\bar{T} = 0.5(3000 + 298) = 1649 \text{ K} \sim 1650 \text{ K}$$

$$\bar{C}_{p,CO_2,1650K} = 59.08 \text{ kJ/kmol} \cdot K$$

$$\bar{C}_{p,H_2O,1650K} = 48.47 \text{ kJ/kmol} \cdot K$$

$$\bar{C}_{p,N_2,1650K} = 35.23 \text{ kJ/kmol} \cdot K$$

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T (°K)	Cp (kJ/kmol·K)		
	CO ₂	H ₂ O	N ₂
298	37.2	33.45	29.07
1000	54.36	41.31	32.76
1100	55.53	42.64	33.36
1200	56.2	43.87	33.71
1300	56.98	45.02	34.11
1400	57.67	46.1	34.48
1500	58.29	47.1	34.8
1600	58.83	48.03	35.1
1700	59.31	48.9	35.36
1800	59.73	49.7	35.59
1900	60.11	50.45	35.8
2000	60.43	51.14	35.99
2100	60.71	51.78	36.15
2200	60.96	52.38	36.3
2300	61.18	52.92	36.43
2400	61.37	53.43	36.54
2500	61.55	53.9	36.64
2600	61.7	54.34	36.74
2700	61.84	54.74	36.82
2800	61.96	55.11	36.89
2900	62.08	55.46	36.96
3000	62.19	55.78	37.03

1

$$T_{ad,guess} = 3\,000^\circ\text{K}$$

$$\bar{T} = 1649\text{ K} \sim 1650\text{ K}$$

$$T_{ad} = 2308.14\text{ K}$$

$$\bar{C}_{p_{\text{CO}_2,1650\text{K}}} = 59.08\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{H}_2\text{O},1650\text{K}}} = 48.47\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{N}_2,1650\text{K}}} = 35.23\text{ kJ/kmol} \cdot \text{K}$$

2

$$T_{ad,guess} = 2308.14^\circ\text{K}$$

$$\bar{T} = 1303.14\text{ K} \sim 1300\text{ K}$$

$$T_{ad} = 2357.48\text{ K}$$

$$\bar{C}_{p_{\text{CO}_2,1300\text{K}}} = 56.98\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{H}_2\text{O},1300\text{K}}} = 45.02\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{N}_2,1300\text{K}}} = 34.11\text{ kJ/kmol} \cdot \text{K}$$

3

$$T_{ad,guess} = 2357.48^\circ\text{K}$$

$$\bar{T} = 1327.81\text{ K} \sim 1330\text{ K}$$

$$\bar{T} \sim 1350\text{ K}$$

$$T_{ad} = 2350.96\text{ K}$$

$$\bar{C}_{p_{\text{CO}_2,1350\text{K}}} = 57.34\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{H}_2\text{O},1350\text{K}}} = 45.57\text{ kJ/kmol} \cdot \text{K}$$

$$\bar{C}_{p_{\text{N}_2,1350\text{K}}} = 34.3\text{ kJ/kmol} \cdot \text{K}$$

Calculate new

T_{ad}:

$$T_{ad} = 298.15\text{ K} + \frac{-249950 - (-5324540) + 0\text{ [kJ/kmol]}}{(8 * 59.08 + 9 * 48.47 + 47 * 35.23)\text{ [kJ/kmol} \cdot \text{K]}} = 2308.14\text{ K}$$

$$T_{ad,guess} \approx T_{ad,new} \quad 3000\text{ K} \gg 2308.14\text{ K}$$

Method 3: C_{p_i} functions

- › Requires to know the constants which dictate the change for a certain compound as a function of temperature
- › As the previous method we will approximate the sensible enthalpy term of the products in order to obtain T_{ad} .

$$\sum N_p (\bar{h}_{f_p}^\circ + \int_{T^\circ}^{T_{ad}} \bar{C}_{p_p} dT) = \sum N_R (\bar{h}_{f_R}^\circ + \int_{T^\circ}^{T_R} \bar{C}_{p_R} dT)$$

Where

$$\bar{C}_{p_i} = a_i + b_i T + c_i T^2 + d_i T^3$$

- › We will look to solve for our unknown variable, T_{ad} .

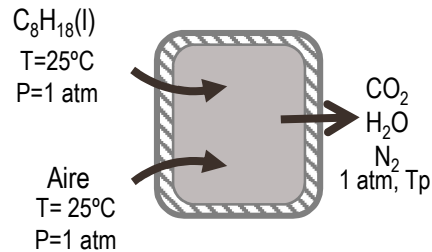
$$T_{ad} = \frac{\sum N_R (\bar{h}_{f_R}^\circ + \int_{T^\circ}^{T_R} \bar{C}_{p_R} dT) - [\sum N_p (\bar{h}_{f_p}^\circ - \bar{C}_{p_p} T^\circ)]}{\sum N_p \bar{C}_{p_p} T_{guess}}$$

- › As with the previous methods, this method will require us to make a supposition or first guess for the value of T_{ad} .

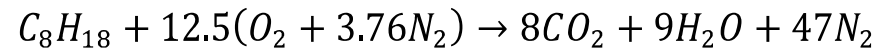
- > Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm

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Suppositions: 1) open system, steady-state; 2) No work is done; 3) All gases are ideal gases



Chemical reaction balanced:



Based on the energy balance:

$$\sum N_P (\cancel{\bar{h}_{f_P}} + \int_{T^\circ}^{T_{ad}} \bar{c}_{p_P} dT) = \sum N_R (\cancel{\bar{h}_{f_R}} + \int_{T^\circ}^{T_R} \bar{c}_{p_R} dT)$$

$$\begin{aligned} \sum N_R (\bar{h}_f^\circ)_R &= -249950 \text{ kJ/kmol} \\ \sum N_P (\bar{h}_f^\circ)_P &= -5\,324\,540 \text{ kJ/kmol} \\ \sum N_R \int_{T^\circ}^{T_R} \bar{c}_{p_R} dT &= 0 \text{ kJ/kmol} \end{aligned}$$

The sensible enthalpy term of the products can be interpreted:

$$\sum N_P \left(\int_{T^\circ}^{T_{ad}} \bar{c}_{p_P} dT \right) = \sum N_P (\bar{c}_{p_P} T_{ad} - \bar{c}_{p_P} T^\circ)$$

Calculating C_{p_P} @ T :

1. Suppose a T_{ad} value
2. Obtain T_{cp}
3. Obtain C_p values @ T_{cp}
Using the C_p equations

$$T_{ad, \text{guess}} = 3\,000^\circ\text{K}$$

$$\bar{T}_{cp} = 0.5(3000 + 298) = 1649 \text{ K}$$



	Ai	Bi	Ci	Di
CO ₂	22.26	5.9810E-02	-3.5010E-05	7.4690E-09
H ₂ O	32.24	1.9230E-03	1.0550E-05	-3.5950E-09
N ₂	28.9	-1.5710E-03	8.0810E-06	-2.8730E-09

$$\overline{Cp}_{CO_2,1649K} = 59.1782 \text{ kJ/kmol} \cdot K$$

$$\overline{Cp}_{H_2O,1649K} = 47.9788 \text{ kJ/kmol} \cdot K$$

$$\overline{Cp}_{N_2,1649K} = 35.4009 \text{ kJ/kmol} \cdot K$$

$$\sum N_p \left(\int_{T^\circ}^{T_{ad}} \overline{Cp}_p dT \right) = \sum N_p (\overline{Cp}_p T_{ad} - \overline{Cp}_p T^\circ)$$

$$= (N_{CO_2} \overline{Cp}_{CO_2} + N_{H_2O} \overline{Cp}_{H_2O} + N_{N_2} \overline{Cp}_{N_2}) T_{ad} - (N_{CO_2} \overline{Cp}_{CO_2} + N_{H_2O} \overline{Cp}_{H_2O} + N_{N_2} \overline{Cp}_{N_2}) T^\circ$$

The values of Cp calculated previously are the ones used for this calculations, and we solve for Tad:

$$T_{ad} = 2273.26 \text{ K} \quad \ll T_{guess} = 3000 \text{ K}$$

The calculated Tad of 2273K becomes our new guessed temperature
And we repeat the process until the value stabilizes

iteration	Tguess	Tcp	Tad	%e
1	3000	1649	2273.26	31.96
2	2273.26	1285.62	2359.11	3.776
3	2359.11	1328.55	2345.57	0.573
4	2345.57	1321.78	2347.65	0.088
5	2347.65	1322.85	2347.33	0.013
6	2347.33	1322.66	2347.38	0.0021

For this specific process the adiabatic flame temperature is 2 347.3 K

Determining T_{ad} – Closed systems

- › All mentioned methods can be used for closed systems
 - Since we suppose all gases behave as ideal gases
 - All internal energy and enthalpies depend on temperature only

$$\sum N_P (\bar{h}_f^\circ + \bar{h}(T_{ad}) - \bar{h}^\circ - P\bar{v})_P = \sum N_R (\bar{h}_f^\circ + \bar{h}(T_F) - \bar{h}^\circ - P\bar{v})_R$$